]Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala

Department of Computer Science and Engineering

SUMMER EST EXAMINATION

Course Code: UTA018	Course Name: Object Oriented Programming
M. Marks: 40	Name of Faculty: Dr. Ashima Khosla
Date: 22nd July, 2025	Time: 2:00 pm - 5:00 pm

Note: Attempt any FOUR questions in scquence. Answer all the sub-parts at one place. Do mention page no. of your attempt at front page of answer sheet. Assume missing data (if any).

No	Question	CO	BL
Q1	I. Explain the following concepts of object-oriented programming: **(2.5 * 2=5 marks)** a) Abstraction b) Inline functions	1	1
	II. Design a C++ program to demonstrate runtime polymorphism using virtual functions and base class pointers. **(5 marks)** 1. Create a base class Polygon with two protected data members of type double: width and height. 2. Include a public method set_value(double, double) to assign values to width and height. 3. Declare a virtual function calculate_area() in the base class. 4. Derive two classes Rectangle and Triangle from Polygon. 5. In both derived classes, override the calculate_area() method: • For Rectangle, compute and display: Area = width × height • For Triangle, compute and display: Area = 0.5 × width × height 6. In the main() function: • Create a base class pointer. • Assign it to objects of Rectangle and Triangle, one at a time. • Use the pointer to call set_value() and calculate_area() to demonstrate polymorphism.	3	6
Q2	I. Write a program in C++ to create a class matrix and overload * operator using friend function to multiply two matrices. **(5 marks)**	3	6
	II. Explain the following C++ concepts with suitable code examples: **(2.5 * 2 = 5 marks)** a) **Multiple Inheritance** – Describe how a class can inherit from more than one base class. Explain possible issues like ambiguity and how they can be resolved. b) **Static Member Functions** – Explain the purpose and usage of static functions in a class. How do they differ from regular member functions?	1	2
Q3	Write a C++ program to create a class named 'Rectangle' with two data members- length and breadth and a function to calculate the area which is 'length*breadth'. The class has three constructors which are: **(10 marks)** • having no parameter - values of both length and breadth are assigned zero. • having two numbers as parameters - the two numbers are assigned as length and breadth respectively. • having one number as parameter - both length and breadth are assigned that number. Now, create objects of the 'Rectangle' class having none, one and two parameters and print their areas.	2	3

Q4	I.	Write a C++ program to accept marks for 5 students. Each mark must be between 0 and 100. If not, throw and catch an exception with an appropriate message. **(5 marks)**	5	3
	II.	Write a C++ program on function template to swap two values. Ensure that both values are of the same data type. **(5 marks)**		
Q5	I.	Explain in brief the purpose of each of the following modes: **(4 marks)** a) ios::app b) ios::binary c) tellp() d) seekp()	4	2 3
	II.	Write a C++ program to Count the number of characters in the file. **(6 marks)**		

**********ALL THE BEST**********